

Commitment To Cooperation With Self-Negotiated Contracts

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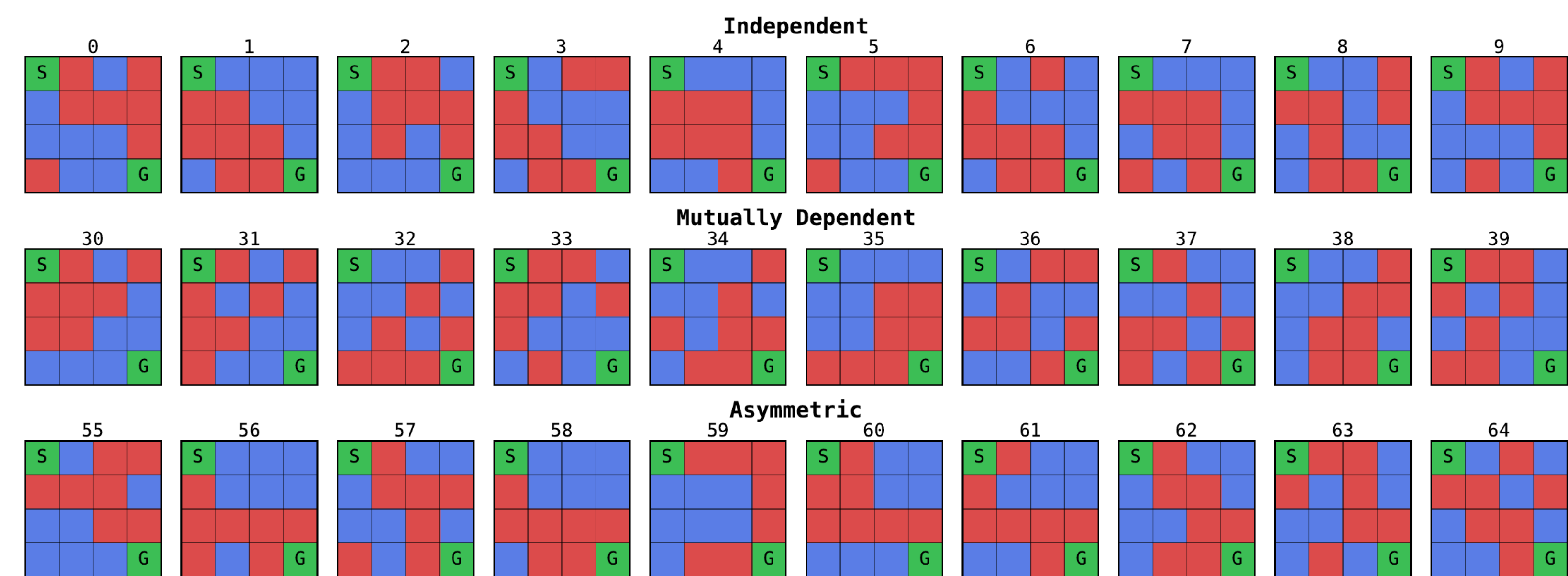
Why Contracts?

Cooperation is hard: its **costs come early, its benefits come later** — an incentive to defect. Human societies solve this with **contracts**: observable agreements whose enforced terms make commitments credible.

Can LLM agents use self-negotiated contracts to cooperate with commitment?

Board Types (80 boards)

- Independent (20)**: each player can finish alone
- Mutually Dependent (20)**: neither can finish without trading
- Asymmetric (40)**: **P-Red** can finish alone, **P-Blue** cannot



Metrics: Normalized Joint Reward, Gini, Both Finished, Both Beat Baseline (BBB), Defection. Six models in self-play on all 80 boards.

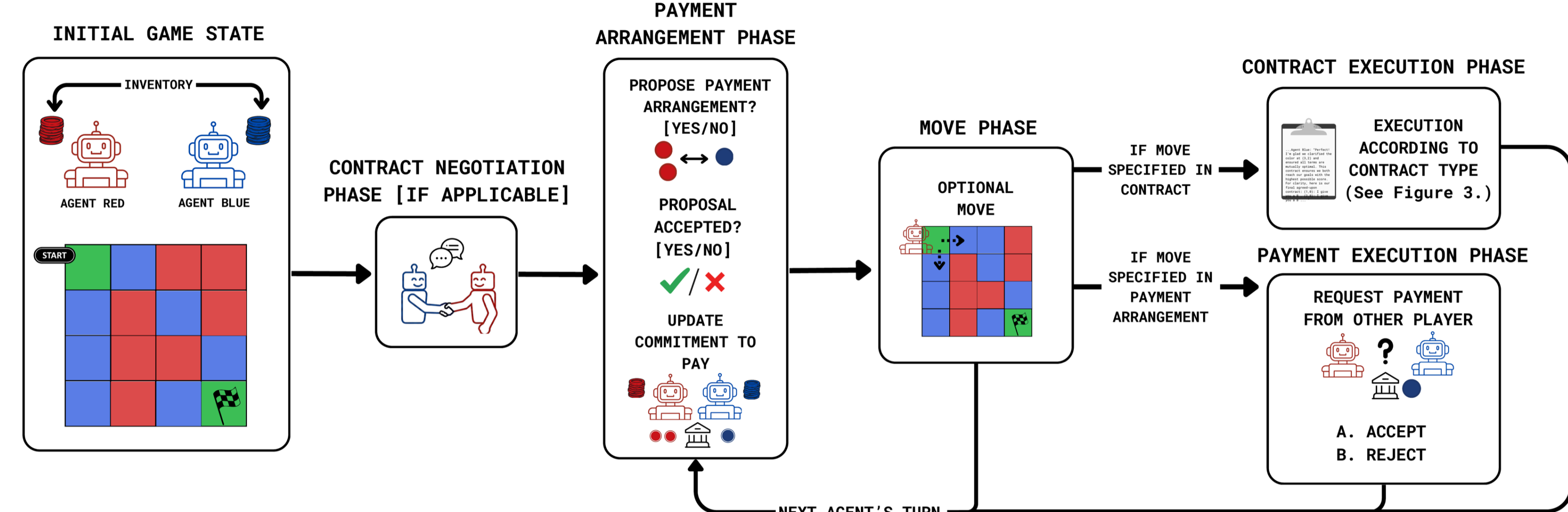
Without Contracts, Cooperation Fails Under Asymmetry

Pay-for-Partner, no contracts. $\hat{R}$ = Normalized Joint Reward; BF = Both Finished; Def. = Defection.

Model	$\hat{R}$	Gini	BF	BBB	Def.
GPT-4.1	0.72	0.28	0.43	0.00	0.23
Haiku-4.5	0.91	0.10	0.82	0.00	0.01
LLaMA Maverick	0.97	0.05	0.95	0.00	0.13
LLaMA Scout	0.84	0.15	0.71	0.00	0.53
Qwen-3-235B	0.66	0.23	0.44	0.00	0.39
Qwen-3-30B	0.52	0.26	0.26	0.00	0.94
Mean	0.77	0.18	0.60	0.00	0.32

BBB = 0.00 for every model: no model got both players above baseline on any Asymmetric board.

The CT-Bench Environment



Game

P-Red and **P-Blue** on a 4x4 board; moving onto a tile costs a chip of its color

Chips

14 chips of own color + 2 green each — crossing the other color requires trades

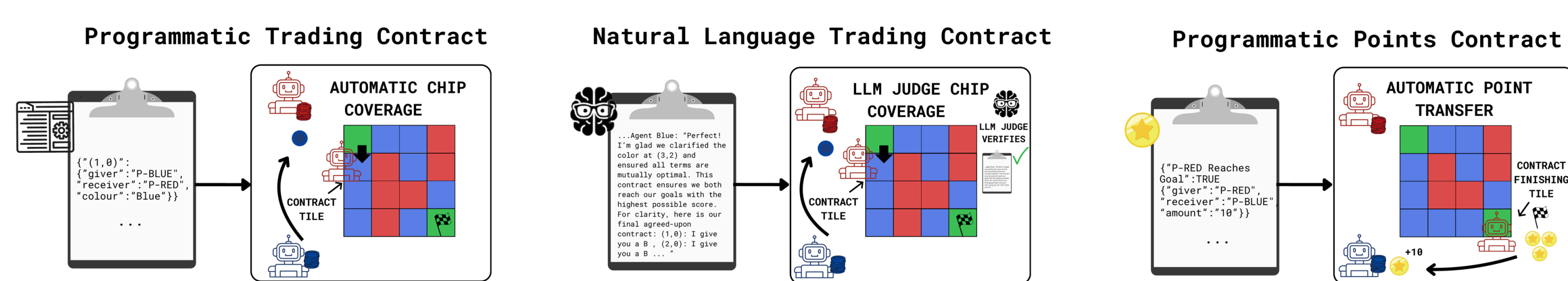
Scoring

20 points for the goal + 5 per remaining chip; each player maximizes its own score

Pay-for-Partner

Players promise to cover each other's moves; promises are **non-binding**

Self-Negotiated Contracts



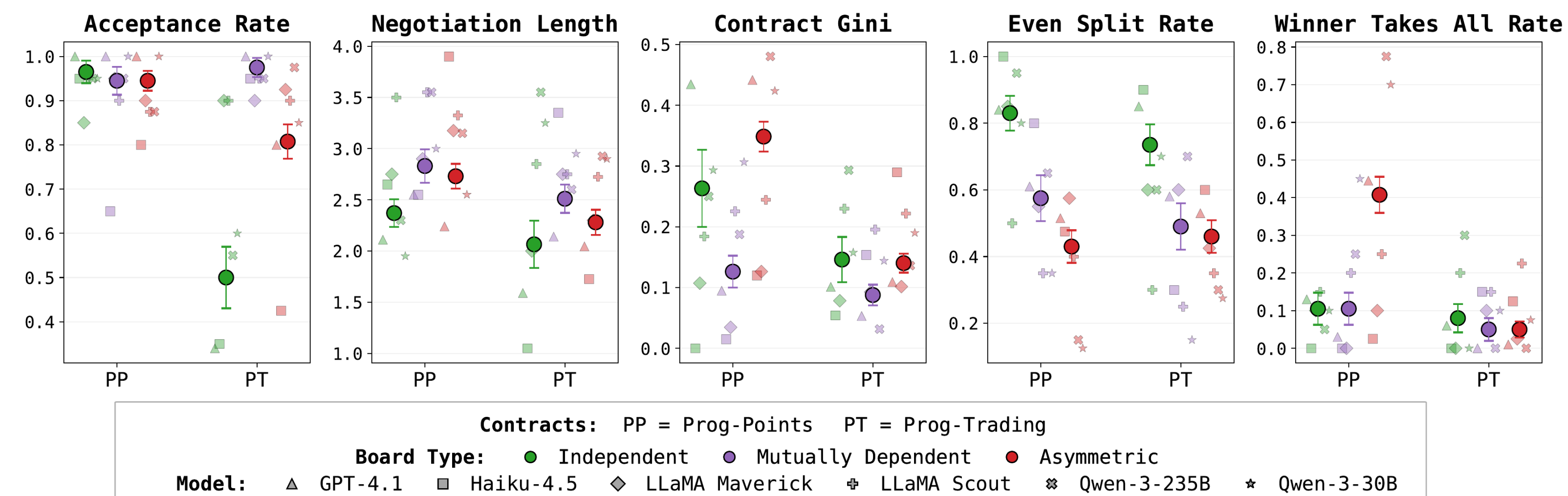
Agents negotiate their own contracts, in three forms:

- Programmatic Trading**: chip transfers to cover agreed tiles, auto-executed
- Natural Language Trading**: contract in natural language, an LLM judge enforces it
- Programmatic Points**: promise part of your final score, auto-paid

```
{ "(row,col)": { "giver": "Player X",  
  "receiver": "Player Y", "color": "<Color>", ... } }
```

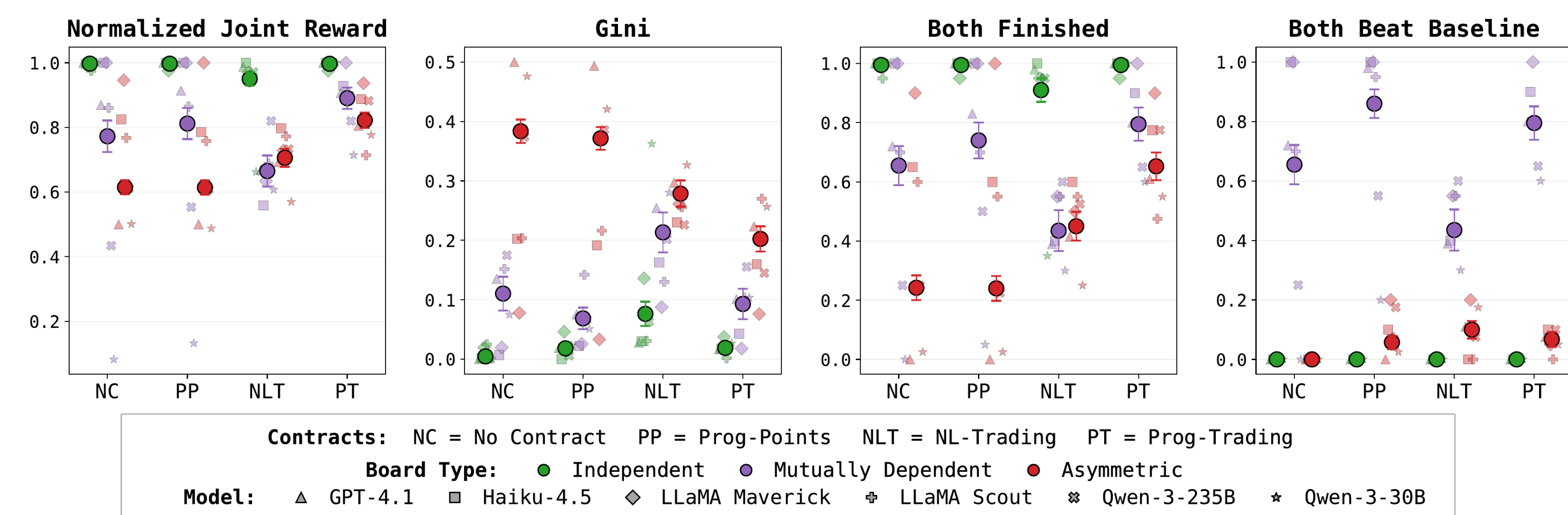
Up to 8 negotiation turns; a judge model (Qwen-3-235B) writes the JSON contract, which both players must accept.

Contract Formation



- Agents reliably agree: **86%** mean acceptance (Haiku-4.5: 67%)
- Representation shapes fairness**: Programmatic Points is winner-takes-all on 39% of Asymmetric boards (Qwen: 74%); Programmatic Trading splits tiles equally in 49%

Contracts Improve Cooperative Outcomes



0.77 → 0.89

Joint Reward, No-Contract → Programmatic Trading ($p < 0.001$)

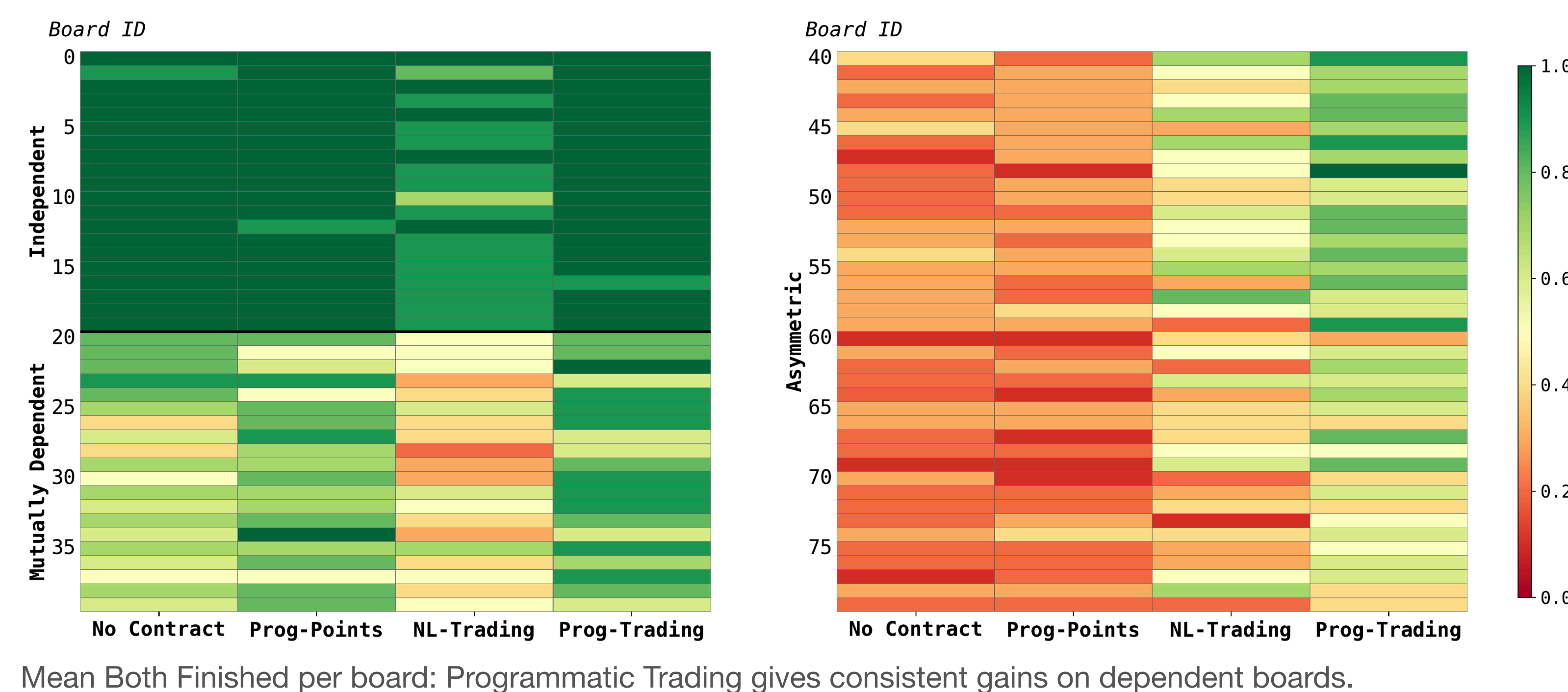
-38%

fewer non-binding promise arrangements

45 vs 6

of 60 dependent boards: best Both Finished

Both Finished, Board by Board



Mean Both Finished per board: Programmatic Trading gives consistent gains on dependent boards.

Contracts Help the Weaker Player

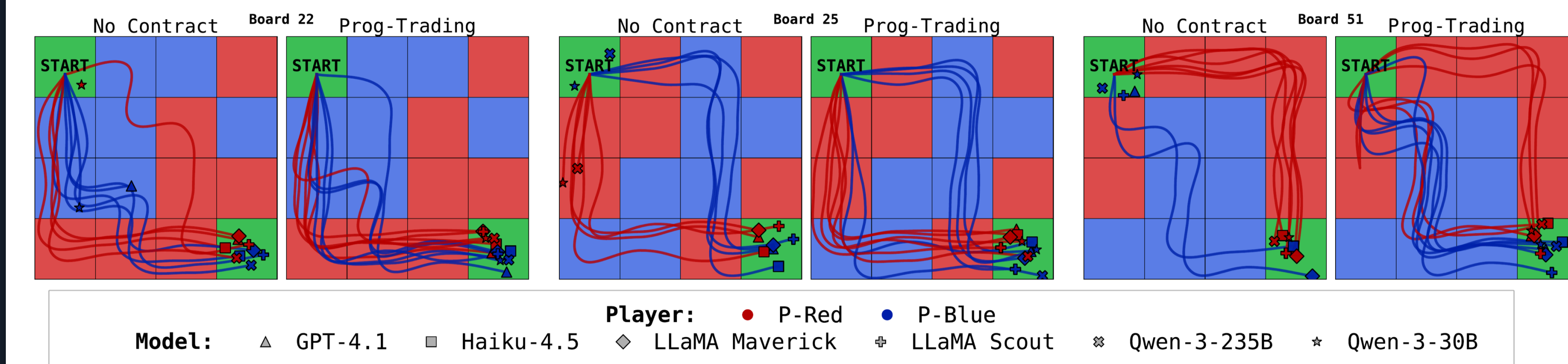
On Asymmetric boards, **P-Blue**'s reward is **68% higher** under Programmatic Trading ($p < 0.001$); **P-Red** gains nothing — the stronger player fails to exploit its position.

When Contracts Backfire

On Mutually Dependent boards, natural-language contracts **underperform no contract at all**: Both Finished 46% vs 61% (No-Contract) vs 79% (Programmatic Trading).

Over-anchoring: contracts formed are similar, but Natural Language Trading suppresses the follow-up trading still needed (1.4 vs 3.0 proposals per game); models declining to trade cite the contract 94% vs 74% of the time.

Contracts Change Behavior on the Board



Model: Δ GPT-4.1 $\square$ Haiku-4.5 $\diamond$ LLaMA Maverick $\star$ LLaMA Scout $\circ$ Qwen-3-235B $\times$ Qwen-3-30B

Player: $\bullet$ P-Red $\bullet$ P-Blue

Contracts can improve outcomes. On boards where models fail to reach their goals without a contract, adding a Programmatic Trading contract lets every model complete its route.

Beyond Self-Play

Pairing strong with weak models on Mutually Dependent boards: under Programmatic Trading the strong model lifts the weak one at its own expense — **cooperation survives large capability gaps**. Without contracts, GPT-4.1 scores 0 against Qwen-3-30B, which honors only **6%** of its promises (Haiku-4.5: 100%).

Takeaways

- The way contracts are represented and enforced matters**: contracts that compile to programs consistently outperform natural-language ones
- Self-negotiated contracts can improve cooperative outcomes** beyond regular trading
- Agents struggle to reach mutually beneficial outcomes from asymmetric starting points**

Paper & Acknowledgments



CT-Bench is open-sourced.
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